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Metrics Monitoring and Alerting System

Step 1 - Understand the Problem and Establish Design Scope

Metrics monitoring and alerting system covers a large infrastructure. We can build all components from scratch. Alternatively, we can build on top of existing data ingestion, storage, and visualization components. Different companies have different requirements and need different systems to fit their needs. At the high level, the following components are needed for a metrics monitoring and alerting system:

Collection: collect metrics data from various sources.
Storage: store the incoming metrics data.
Alerting: receive alerts from other components and notify various communication channels.
Visualization: show metrics and alerts in various forms.

Candidate: Who are the users of the system? Is it for internal use only?
Interviewer: Only for internal use by our employees.
Candidate: How many metrics do you plan to monitor?
Interviewer: Assume it is a large number, we need to be able to scale to support this.
Candidate: Do we need to collect metrics from the operating system?
Interviewer: Yes, we need to collect OS level metrics.
Candidate: Do we need to collect metrics from the database?
Interviewer: Yes, we need to collect metrics from the database.
Candidate: Do we need to collect metrics from distributed cache?
Interviewer: Yes, we need to collect metrics from distributed cache.
Candidate: What is the retention period?
Interviewer: Assume we need to keep data for two weeks.
Candidate: Can we change the resolution of the metrics data for storage?
Interviewer: Yes, this is a good question. We will talk about it later in the interview.
Candidate: What time intervals do we need to support?
Interviewer: Time intervals should be configurable and large enough to support business and operational use cases.
Candidate: Do we need to support alerting?
Interviewer: Yes, we need to support alerting.
Candidate: Do we need to support anomaly detection?
Interviewer: No, we do not need to support anomaly detection.

High level requirements and assumptions

Our in-house monitoring system is large and sophisticated. We only need to support the following:

- Collect metrics data from various sources.
- Store the collected data.
- Query and visualize metrics data.
- Send alerts when a metric violates its specified rule. For example, alert when disk usage reaches 90% or more.
- Support business and operational use cases.
- Support a large infrastructure.
- Support alerting.
- Support data retention.
- Support data resolution.
- Support configurable time intervals.

Non-functional requirements

- Scalability: the system should be scalable to accommodate more metrics and alerts.
- Latency: queries should be processed as fast as possible.
- Reliability: the system should be reliable to avoid missing critical alerts.
- Flexibility: leaving high-level options for extension and integration (such as integrate with a new technology).

Back of the envelope estimation

- Assume 100 million daily active users (DAU).
- Assume a large internet company has 1000 server machines, each machine has 100 metrics to run.
- We need to collect metrics data every one minute.

Step 2 - Propose High-level Design and Get Buy-In

Fundamentals

A simple monitoring system may look like the following:

Figure 21.1 Simple monitoring system

The system contains the following components:

- Time series database (TSDB): a database specifically optimized for time series data.
- Metrics collection: collect metrics data from various sources.
- Metrics storage: a system to store and manage metrics data.
- Alerting: receive alerts from other components and notify various communication channels.
- Visualization: show metrics and alerts in various forms.

Data model

Metrics data is usually recorded in the form of time series. A time series is a series of values which are associated with a timestamp. For example, the following time series represents the instance count of a web server: